



EDUCATION

- Indian Institute of Technology Bombay**, Grade: 8.60/10.00 [2024]
Master of Technology, Computer Integrated Manufacturing
Bachelor of Technology, Mechanical Engineering, Minor in Industrial Engineering & Operations Research (IEOR)
- Eklavya School of Science**, Grade (PCM): 89.50%, Higher Secondary Certificate (Class XII), GHSEB [2018]
- Adarsh Vidhyalaya**, Grade: 91.33%, Secondary School Certificate (Class X), GSEB [2016]

SCHOLASTIC ACHIEVEMENTS & AWARDS

- Attained **AA grades** in **ME781 - Statistical ML & Data Mining** and **GNR638 - ML for Remote Sensing** courses [2022]
- Awarded the **Young Researchers' Prize** by IEEE OES by the **University of Tokyo, Japan** as **AUV-IITB** team member [2021]
- Obtained an All India Rank of **644** in **JEE Mains** (1.15 million appeared) | **97.9 percentile** in **IIT-JEE Advanced** [2019]
- Secured **1st rank** in Gujarat State with an All India Rank of **48** in **JEE-Mains Paper-2 (B Arch./B. Planning)** [2019]
- Obtained **70/70** in **Science** in **SSC (X)** (**99.74 percentile**) and **100/100** score in **Mathematics** in **HSC (XII)** [2016-18]

PROFESSIONAL EXPERIENCE

Software Engineer | *Spectral Insights Pvt Ltd (Pramana), Bangalore* [Jul '24 - Present]

AI-driven health-tech company specializing in digital pathology, building autonomous whole-slide imaging systems and software.

- Camera IQC:** Enhanced camera quality assurance by integrating intensity-based and spatial clustering algorithms for defect classification, ensuring only high-reliability cameras are integrated into whole-slide imaging systems.
- Hazy Edge Correction:** Eliminated **hazy stitching artifacts** in whole-slide images (WSI) by designing an overlap correction pipeline, improving image clarity across diverse slide types and scanner models.
- Even Stack Acquisition:** Redesigned Z-stack acquisition to support both even and odd stack sizes, improving pipeline flexibility and delivering **3.25% faster scans per mm²** with **no loss** in image quality.
- Rescan Optimization:** Improved rescan decision-making by filtering artifact-driven focus errors, achieving a **68.9% reduction** in unnecessary rescans and significantly enhancing tissue-focused image quality.

Data Analytics Intern | *MedTourEasy, New Delhi* [May '23]

*Blood Donations Prediction, Received **appreciation certificate** and **letter of recommendation***

- Developed an ML model using **logistic regression** for identifying patterns and factors influencing blood donation behavior
- Achieved an AUC score of **0.7634** by performing an intelligent search over ML pipelines to select an algorithm using **TPOT**
- Enhanced the AUC score by **3.37%** through the strategic application of **log normalization** to the high-variance features

Technical Intern | *Wings EV, Indore* [Dec '21 - Jan '22]

- Developed a **C++ library** on Raspberry Pi to process CAN bus data, enabling **audio alerts** based on received signal bytes
- Integrated functionality to play audio files through a speaker, enhancing **real-time notification** capabilities for the system

MASTERS' THESIS & OTHER KEY PROJECTS

AI Classifier for Wear and Lubrication State Identification [Jun '23 - Jun '24]

Received AB grade in M-Tech Project | Department of Mechanical Engineering, IIT Bombay

Guide: Prof. Asim Tewari

- Automated **lubrication state monitoring** and **friction force prediction** in industrial machinery, reducing manual checks
- Designed custom **charge amplifier circuits** and **Arduino/ESP-32** systems for **high-frequency vibration** data acquisition
- Applied statistical ML models (RF, XGBoost, Logistic/Linear Regression) with feature engineering, PCA, and scaling
- Achieved **>98%** lubrication state classification accuracy and **R² = 0.83** in friction force prediction, reducing downtime

Virtual Moon Lander | *Machine Learning Specialization | Stanford University* [Feb '23 - Jul '23]

- Designed Rocket Trajectory Optimization **RL-Model** using **TensorFlow** and OpenAI's Environment Rendering Library Gym
- Reduced **network instabilities** and **oscillations** problems using **Experience Replay** and **Soft Weight Update** techniques
- Solved RL-Environment for absolute precision safety landing, achieving a **threshold score** of **200+** in just **534 Episodes**

Matsya, Autonomous Underwater Vehicle (AUV) [Oct '19 - Jul '22]

RoboSub, RoboNation

Project Guide: Prof. Leena Vachhani, Prof. Hemendra Arya

*All student team working on the design & development of a state-of-the-art AUV, **Matsya**, which competes at **RoboSub, USA***

Accolades: IEEE Young Researchers' Prize awardee, University of Tokyo; 3rd prize in RoboSub 2020; Featured in Janes

Electrical Subdivision

- Designed **Power Board** in EAGLE which has **12 independent sensors** toggle lines, **reverse-voltage protection**, Hard-kill for thrusters, the ability to generate required multiple voltages across the vehicle and measure the voltage of a battery
- Developed the code for the microcontroller performing functions like **I/O signals**, Power toggles, **UART** communication

TECHNICAL SKILLS & CERTIFICATIONS

C++, Python, R, SQL, MongoDB, Keras, Scikit-Learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Tableau, MS Excel, IBM Cognos, MATLAB, Git/GitHub | **Certifications:** Google Data Analytics, IBM Data Science, Stanford ML Specialization